# A Period Formula for Rational Cut-and-Project Strip Projections with Multiplicity

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**Abstract**

Consider the one-dimensional cut-and-project construction with coprime positive integers $\alpha, \beta$ defining the rational slope $\alpha/\beta$, and window parameter $\omega \geq 0$. Projecting lattice points from the resulting strip onto the line and counting them with multiplicity gives a multiset whose consecutive gaps form a periodic sequence. We prove that its minimal period is

$$\lambda = \begin{cases} N, & D \nmid N, \\ N/D, & D \mid N, \end{cases}$$

where

$$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D = \alpha^2 + \beta^2.$$

The proof reduces the geometry to the distribution of $N$ consecutive residue classes modulo $D$. When this distribution is non-uniform, the residue classes with larger multiplicity form a proper cyclic interval in the natural residue parametrisation; its only translational symmetry is the identity, which rules out any smaller period. We also discuss the effect of discarding multiplicities, in particular in the uniform case $D \mid N$. The algebraic core of the proof has been formalised in Lean 4.

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**Keywords:** cut-and-project sets, projected multisets, gap sequences, period length, rational slope, residue classes, lattice projection, Lean 4 formalisation.

## 1 Introduction

Cut-and-project constructions provide a standard way of producing ordered point sets from lattices. In the one-dimensional planar case, one selects lattice points lying in a strip around a line and projects them onto that line. When the slope is irrational, the resulting objects are typically aperiodic and belong to the classical theory of model sets and aperiodic order. When the slope is rational, the resulting structure is periodic and is therefore often regarded as a degenerate or elementary case from the aperiodic-order point of view.

This viewpoint goes back in part to Yves Meyer's foundational work on harmonious sets and model sets [8]. Meyer's work became one of the mathematical foundations of the modern theory of quasicrystals and aperiodic order, and was recognised by the 2017 Abel Prize.

Nevertheless, even in the rational case there are natural quantitative questions. One such question is the following: after projecting the accepted lattice points and ordering the projected

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values along the line, what is the minimal period of the sequence of consecutive gaps? This question depends on a convention. If projected values are counted with multiplicity, then distinct lattice points projecting to the same value give rise to zero gaps. If multiplicities are discarded, the resulting ordinary point-set gap sequence may have a smaller period. The main theorem of this paper is stated for the projected multiset.

Let the rational slope be $\alpha/\beta$, with $\gcd(\alpha, \beta) = 1$, and let $\omega \geq 0$ be the window parameter. We prove that the multiset gap sequence has minimal period

$$\lambda = \begin{cases} N, & D \nmid N, \\ N/D, & D \mid N, \end{cases}$$

where

$$N = \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D = \alpha^2 + \beta^2.$$

Here $N$ is the number of admissible values of the invariant $r = \alpha x - \beta y$, and $D$ is the common difference of the arithmetic progressions obtained after projection.

The proof is elementary. The accepted lattice points decompose into parallel tracks indexed by the admissible values of $r$. Each track gives an arithmetic progression of projected values with common difference $D$, and the relevant residues modulo $D$ are obtained from $N$ consecutive integers by multiplication with a unit modulo $D$. If $D \nmid N$, the residue classes with larger multiplicity form a proper cyclic interval in the natural residue parametrisation; its only translational symmetry is the identity, which rules out any smaller period. If $D \mid N$, the residues are uniformly distributed and the gap sequence collapses to a simple pattern of period $N/D$.

The algebraic core of the proof has been formalised in Lean 4. The formalisation verifies the residue-class and minimal-period arguments; the geometric construction is represented at the combinatorial level described above.

The paper is organised as follows. Section 2 gives the construction and fixes the multiplicity convention. Section 3 states the main result. Section 4 proves the generic case $D \nmid N$. Section 5 treats the uniform case $D \mid N$. Section 6 records some immediate corollaries. Section 7 describes the Lean 4 formalisation. Section 8 discusses related literature. Section 9 discusses the set-valued variant and open questions.

# 2 Construction of the Sequences

## 2.1 Accepted lattice points

Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and let $\omega \in \mathbb{R}_{\geq 0}$. Consider the line $f(x) = \frac{\alpha}{\beta} x$ in $\mathbb{R}^2$ together with the strip bounded by

$$l(x) = \frac{\alpha}{\beta}(x - \omega) \qquad \text{and} \qquad u(x) = \frac{\alpha}{\beta} x + \omega.$$

The set of *accepted lattice points* is

$$\mathbf{C} \;=\; \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^2 \;\middle|\; y \in \left[ \frac{\alpha}{\beta}(x - \omega),\; \frac{\alpha}{\beta} x + \omega \right] \right\}. \tag{1}$$

## 2.2 Projection and ordering

The orthogonal projection of $(x, y) \in \mathbb{R}^2$ onto the line $f$ is

$$\mathbf{p} \;=\; \frac{1}{\alpha^2 + \beta^2} \begin{pmatrix} \beta^2 & \alpha\beta \\ \alpha\beta & \alpha^2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{2}$$

The $x$-coordinate of the projected point is

$$p_x = \frac{\beta(\beta x + \alpha y)}{\alpha^2 + \beta^2}.$$

Since the factor $\frac{\beta}{\alpha^2+\beta^2}$ is a positive constant, the ordering of projected points is determined by

$$\tilde{p} \ := \ \beta x + \alpha y. \tag{3}$$

### 2.3 Multiplicity convention

Throughout this paper, projected values are counted with multiplicity. Thus, if two distinct accepted lattice points have the same projected coordinate, they contribute two consecutive entries to the sorted projected sequence and may therefore produce a zero gap.

Equivalently, we study the gap sequence of the projected lattice-point *multiset*, not only the underlying projected point set. If multiplicities are discarded, the resulting ordinary point-set gap sequence may have a smaller period. In particular, in the case $D \mid N$, the set-valued projected sequence has constant gap and period 1, whereas the multiset sequence has period $N/D$.

### 2.4 Distance sequence and period

As shown below in Section 4.2, the projected values form a finite union of arithmetic progressions with common difference $\alpha^2+\beta^2$, counted with finite multiplicities. Hence the resulting multiset is locally finite and unbounded in both directions. It therefore admits a nondecreasing enumeration indexed by $\mathbb{Z}$, unique up to translation of the index.

Sort the multiset of all $\tilde{p}$-values in nondecreasing order, with multiplicities retained, to obtain $(\tilde{p}_s^{(i)})_{i\in\mathbb{Z}}$, and define the *difference sequence*

$$\delta^{(i)} \ := \ \tilde{p}_s^{(i+1)} - \tilde{p}_s^{(i)}. \tag{4}$$

Since orthogonal projection onto the line rescales the ordered $\tilde{p}$-coordinate by a positive constant, the period length of $(\delta^{(i)})$ equals the period length of the corresponding Euclidean distance sequence between consecutive projected multiset entries, including zero distances arising from coincident projected points.

**Definition 2.1.** The sequence $(\delta^{(i)})_{i\in\mathbb{Z}}$ has *period length* $\lambda \in \mathbb{N}$ if $\lambda$ is the smallest positive integer such that $\delta^{(i+\lambda)} = \delta^{(i)}$ for all $i$.

### 2.5 The trivial case $\omega = 0$

When $\omega = 0$, the only accepted lattice points lie on $f$ itself: $(x, y) = k(\beta, \alpha)$ for $k \in \mathbb{Z}$. Then

$$\tilde{p} = \beta \cdot k\beta + \alpha \cdot k\alpha = k(\alpha^2 + \beta^2),$$

so all differences equal $\alpha^2 + \beta^2$ and $\lambda = 1$. Note that the formula (6) agrees: $N = 0 + 0 + 1 = 1$ and $D = \alpha^2 + \beta^2 \geq 2$, so $D \nmid 1$ and $\lambda = N = 1$.

## 3 Main Result

**Theorem 3.1** (Period length formula). *Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and let $\omega \in \mathbb{R}_{\geq 0}$. Define*

$$N \ := \ \lfloor \omega\alpha \rfloor + \lfloor \omega\beta \rfloor + 1, \qquad D \ := \ \alpha^2 + \beta^2. \tag{5}$$

*Then the period length of the difference sequence (4) is*

$$\lambda_{\alpha,\beta} \ = \ \begin{cases} N & \text{if } D \nmid N, \\ N/D & \text{if } D \mid N. \end{cases} \tag{6}$$

*Remark* 3.2 (Interpretation of $N$). The quantity $N$ equals the number of integers in the interval $[-\beta\omega,\ \alpha\omega]$, which is the range of the invariant $r = \alpha x - \beta y$ (see Section 4.1). Geometrically, $N$ counts the number of tracks, namely the parallel arithmetic progressions of projected values counted with multiplicity.

*Remark* 3.3 (Interpretation of $D$). The quantity $D = \alpha^2 + \beta^2$ is the common difference of each arithmetic progression of $\tilde{p}$-values. It also equals $\|(\beta, \alpha)\|^2$, the squared length of the direction vector of $f$.

*Remark* 3.4 (Computational verification). The formula has been verified against:

- 51,012 test cases from [4] with $\alpha, \beta$ up to $\sim$100,000 and rational $\omega$ up to $\sim$10, all satisfying $D \nmid N$ (51,011 with $N < D$, one with $N \geq D$);
- 468 deliberately constructed cases in the degenerate regime $D \mid N$ (with $\alpha, \beta \leq 10$ and $N$ up to $\sim$500), verifying the $\lambda = N/D$ branch.

All 51,480 cases produce exact matches with zero failures.

# 4 Proof: The Generic Case $D \nmid N$

## 4.1 Step 1: Range of the invariant

The strip condition (1), multiplied by $\beta > 0$, reads

$$\alpha(x - \omega) \ \leq\ \beta y \ \leq\ \alpha x + \beta\omega,$$

equivalently

$$-\beta\omega \ \leq\ \underbrace{\alpha x - \beta y}_{=:\, r} \ \leq\ \alpha\omega. \tag{7}$$

Since $r \in \mathbb{Z}$, it takes values in

$$\mathcal{R} \ :=\ \big\{-\lfloor\beta\omega\rfloor,\ -\lfloor\beta\omega\rfloor + 1,\ \ldots,\ \lfloor\alpha\omega\rfloor\big\}, \tag{8}$$

using the identity $\lceil -x \rceil = -\lfloor x \rfloor$. Hence

$$|\mathcal{R}| = \lfloor\alpha\omega\rfloor + \lfloor\beta\omega\rfloor + 1 = N.$$

The invariant $r$ is constant along each track: if $(x, y)$ and $(x + \beta, y + \alpha)$ are both accepted, they share the same $r$, since

$$\alpha(x + \beta) - \beta(y + \alpha) = \alpha x - \beta y.$$

The step $(\beta, \alpha)$ is the direction of $f$.

## 4.2 Step 2: Each value of $r$ yields one arithmetic progression

For a fixed $r \in \mathcal{R}$, the system

$$\beta x + \alpha y = \tilde{p}, \qquad \alpha x - \beta y = r \tag{9}$$

has determinant $-(\alpha^2 + \beta^2) = -D \neq 0$, giving the unique solution

$$x = \frac{\beta\tilde{p} + \alpha r}{D}, \qquad y = \frac{\alpha\tilde{p} - \beta r}{D}. \tag{10}$$

Integrality requires $D \mid (\beta\tilde{p} + \alpha r)$, i.e.

$$\tilde{p} \ \equiv\ c_r \ :=\ -\alpha\beta^{-1} r \pmod{D}, \tag{11}$$

where $\beta^{-1}$ is the modular inverse of $\beta$ modulo $D$ (existence: Corollary 4.2). The set of $\tilde{p}$-values for invariant $r$ is the arithmetic progression

$$P_r \;=\; \{c_r + kD : k \in \mathbb{Z}\}. \tag{12}$$

The complete multiset of projected values is

$$S = \bigcup_{r \in \mathcal{R}} P_r,$$

where the union is understood as a multiset union.

### 4.3 Step 3: Coprimality lemmas

**Lemma 4.1.** $\gcd(\alpha, D) = \gcd(\beta, D) = 1$.

*Proof.* $D = \alpha^2 + \beta^2 \equiv \beta^2 \pmod{\alpha}$, so any common divisor of $\alpha$ and $D$ divides $\beta^2$. But $\gcd(\alpha, \beta) = 1$ implies $\gcd(\alpha, \beta^2) = 1$. The argument for $\beta$ is symmetric. □

**Corollary 4.2.** *The modular inverse $\beta^{-1} \pmod{D}$ exists, and the map $r \mapsto c_r = -\alpha\beta^{-1} r \bmod D$ is a bijection on $\mathbb{Z}/D\mathbb{Z}$.*

### 4.4 Step 4: Structure of residues

The $N$ residues $\{c_r : r \in \mathcal{R}\}$ are the image of $N$ consecutive integers under multiplication by $m := -\alpha\beta^{-1} \bmod D$, with $\gcd(m, D) = 1$. Explicitly:

$$\{c_r\}_{r \in \mathcal{R}} \;=\; \{m \cdot r_0,\ m(r_0 + 1),\ \ldots,\ m(r_0 + N - 1)\} \bmod D, \tag{13}$$

where $r_0 = -\lfloor \beta\omega \rfloor$.

**Lemma 4.3** (Non-uniform residue distribution)**.** *If $D \nmid N$, the multiset of $N$ residues $\{c_r \bmod D : r \in \mathcal{R}\}$ is not uniformly distributed over $\mathbb{Z}/D\mathbb{Z}$. More precisely, writing $N = qD + s$ with $0 < s < D$, exactly $s$ residue classes are hit $q + 1$ times and $D - s$ residue classes are hit $q$ times.*

*Proof.* The map $r \mapsto mr \bmod D$ is a bijection on $\mathbb{Z}/D\mathbb{Z}$ by Corollary 4.2. As $r$ runs through $N$ consecutive integers, the residues $mr \bmod D$ cycle through all $D$ classes with period $D$. After $q$ complete cycles, all classes have been hit exactly $q$ times. The remaining $s = N - qD$ values hit $s$ additional classes exactly once each. Hence $s$ classes have multiplicity $q + 1$ and $D - s$ classes have multiplicity $q$. □

### 4.5 Step 5: Period equals $N$ when $D \nmid N$

*Proof.* Within each interval $[kD, (k + 1)D)$ on the $\tilde{p}$-axis, exactly $N$ points from the multiset $S$ occur, counted with multiplicity. Their positions within the interval are determined by the residues $c_r \bmod D$, again counted with multiplicity.

Let

$$0 \le v_0 \le v_1 \le \cdots \le v_{N-1} < D$$

be the nondecreasing list of these residues in one interval of length $D$, with repetitions according to their multiplicities. Then the sorted projected multiset is obtained by repeating this same residue block in every translate by $D$. More explicitly, after a possible translation of the indexing, every $i \in \mathbb{Z}$ can be written uniquely as

$$i = kN + j, \qquad k \in \mathbb{Z}, \quad 0 \le j < N,$$

and

$$\tilde{p}_s^{(i)} = kD + v_j.$$

Consequently,

$$\tilde{p}_s^{(i+N)} = \tilde{p}_s^{(i)} + D$$

for all $i \in \mathbb{Z}$. Therefore

$$\delta^{(i+N)} = \tilde{p}_s^{(i+N+1)} - \tilde{p}_s^{(i+N)} = \tilde{p}_s^{(i+1)} - \tilde{p}_s^{(i)} = \delta^{(i)}$$

for all $i$. Thus $N$ is a period of the difference sequence, and hence the minimal period $\lambda$ divides $N$.

**Minimality.** We now prove that no proper divisor of $N$ is a period, using a multiplicity-function argument on $\mathbb{Z}/D\mathbb{Z}$.

Define the multiplicity function

$$\mu\colon \mathbb{Z}/D\mathbb{Z} \to \mathbb{N}_0$$

by

$$\mu(c) \;:=\; \big|\{r \in \mathcal{R} : c_r \equiv c \pmod{D}\}\big|.$$

By Lemma 4.3, with $N = qD + s$ and $0 < s < D$, the function $\mu$ takes exactly two values: $q+1$ on a set of $s$ residue classes and $q$ on the remaining $D - s$ classes.

It is convenient to pass to the $m$-stepping parametrisation

$$f(x) := \mu(mx), \qquad x \in \mathbb{Z}/D\mathbb{Z},$$

where $m = -\alpha\beta^{-1} \pmod{D}$. Since $m$ is invertible modulo $D$, this is only a relabelling of the same data. In this parametrisation, the heavy set

$$H := \{x \in \mathbb{Z}/D\mathbb{Z} : f(x) = q + 1\}$$

is a contiguous cyclic interval of length $s$:

$$H = \{x_0, x_0 + 1, \ldots, x_0 + s - 1\} \pmod{D}$$

for some $x_0$, because the final $s$ elements of the consecutive block $r_0, r_0 + 1, \ldots, r_0 + N - 1$ contribute one extra hit exactly on $s$ consecutive values in the $m$-stepping order.

Now suppose, for contradiction, that the minimal period $\lambda'$ satisfies $\lambda' < N$. Since $N$ is a period, the minimal period divides $N$. Thus we may write

$$N = k\lambda'$$

with $k > 1$.

Let

$$\sigma := \delta^{(0)} + \delta^{(1)} + \cdots + \delta^{(\lambda'-1)}.$$

Since the gap sequence has period $\lambda'$, we have

$$\tilde{p}_s^{(i+\lambda')} = \tilde{p}_s^{(i)} + \sigma$$

for all $i$. Hence the entire multiset $S$ is invariant under translation by $\sigma$.

On the other hand, $N$ consecutive gaps always sum to $D$, because

$$\tilde{p}_s^{(i+N)} - \tilde{p}_s^{(i)} = D.$$

Since $N = k\lambda'$, this gives

$$k\sigma = D.$$

Thus $\sigma = D/k$ is a positive integer (positive because $D > 0$ and $k \geq 1$) and $k \mid D$. In particular $0 < \sigma < D$, since $k > 1$.

The translation invariance of $S$ implies invariance of the residue multiplicity function:

$$\mu(c+\sigma) = \mu(c) \qquad \text{for all } c \in \mathbb{Z}/D\mathbb{Z}.$$

Equivalently, the relabelled function $f$ satisfies

$$f(x+\tau) = f(x), \qquad \tau := m^{-1}\sigma \pmod{D}.$$

Since $m$ is invertible modulo $D$ and $0 < \sigma < D$, we have $\tau \not\equiv 0 \pmod{D}$. Therefore its heavy set $H$ must satisfy $H + \tau = H$.

But a proper nonempty cyclic interval in $\mathbb{Z}/D\mathbb{Z}$ has trivial stabiliser: if $0 < s < D$ and

$$H = \{x_0, x_0+1, \dots, x_0+s-1\} \pmod{D},$$

then $H + \tau = H$ forces $\tau \equiv 0 \pmod{D}$. Indeed, $x_0 \in H$ while $x_0 - 1 \notin H$, so $x_0$ is the unique element of $H$ whose predecessor is not in $H$. If $H + \tau = H$, then $x_0 + \tau$ must have the same property and hence must equal $x_0$. Thus $\tau \equiv 0 \pmod{D}$.

This contradiction shows that no proper divisor $\lambda' < N$ can be a period. Since $N$ is a period, it follows that the minimal period is

$$\lambda = N.$$

$\square$

# 5 The Degenerate Case $D \mid N$

**Theorem 5.1.** *If $D \mid N$, then $\lambda = N/D$.*

*Proof.* When $D \mid N$, the $N$ values of $r$ span exactly $N/D$ complete cycles of the map $r \mapsto mr \bmod D$. Since this map is a bijection on $\mathbb{Z}/D\mathbb{Z}$, each residue class $c \in \{0, 1, \dots, D-1\}$ is hit exactly $N/D$ times. Consequently, each residue class contributes $N/D$ projected multiset entries per interval of length $D$ on the $\tilde{p}$-axis. By the uniform distribution, every integer $\tilde{p}$ is achieved with multiplicity $N/D$.

The sorted sequence of $\tilde{p}$-values, counted with multiplicity, thus consists of each integer repeated $N/D$ times:

$$\dots,\ \underbrace{t,\ t,\ \dots,\ t}_{N/D},\ \underbrace{t+1,\ t+1,\ \dots,\ t+1}_{N/D},\ \dots$$

The difference sequence is

$$\underbrace{0,\ 0,\ \dots,\ 0}_{N/D-1},\ 1,\ \underbrace{0,\ 0,\ \dots,\ 0}_{N/D-1},\ 1,\ \dots$$

which has period $N/D$. $\square$

*Example* 5.2. For $\alpha = 2$, $\beta = 1$, $\omega = 3$:

$$N = \lfloor 6 \rfloor + \lfloor 3 \rfloor + 1 = 10, \qquad D = 5.$$

Since $5 \mid 10$, the formula gives $\lambda = 10/5 = 2$. Direct computation confirms the difference sequence $(0, 1, 0, 1, \dots)$ with period 2.

*Example* 5.3. For $\alpha = 3$, $\beta = 2$, $\omega = 5$:

$$N = \lfloor 15 \rfloor + \lfloor 10 \rfloor + 1 = 26, \qquad D = 13.$$

Since $13 \mid 26$, the formula gives $\lambda = 26/13 = 2$.

*Remark* 5.4. The boundary case $N = D$, i.e. $N/D = 1$, gives $\lambda = 1$: every integer is a projected value with multiplicity 1, and all gaps equal 1.

## 6 Corollaries

Define $\Lambda_{\alpha,\beta} := \alpha + \beta + 1$.

**Corollary 6.1** (Finiteness)**.** *For all $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$ and all $\omega \in \mathbb{R}_{\geq 0}$, the period length $\lambda$ is finite.*

*Proof.* $N$ is a finite positive integer, and $\lambda \leq N$. □

**Corollary 6.2** (Period at $\omega = 1$)**.** *For $\omega = 1$, $\lambda = \alpha + \beta + 1 = \Lambda_{\alpha,\beta}$.*

*Proof.* $N = \alpha + \beta + 1$ and $D = \alpha^2 + \beta^2$. For $(\alpha, \beta) \neq (1, 1)$, we have

$$\alpha + \beta + 1 < \alpha^2 + \beta^2,$$

so $D \nmid N$ and $\lambda = N = \alpha + \beta + 1$. For $(\alpha, \beta) = (1, 1)$, $N = 3$, $D = 2$, and again $D \nmid N$, so $\lambda = 3 = \Lambda_{1,1}$. □

**Corollary 6.3** ($0 < \omega < 1$)**.** *If $0 < \omega < 1$, then*

$$\lambda < \Lambda_{\alpha,\beta}.$$

*Proof.* $\lfloor \omega\alpha \rfloor \leq \alpha - 1$ and $\lfloor \omega\beta \rfloor \leq \beta - 1$, so

$$N \leq \alpha + \beta - 1 < \Lambda_{\alpha,\beta}.$$

Since $\alpha + \beta - 1 < \alpha^2 + \beta^2 = D$ for all $\alpha, \beta \geq 1$, we have $N < D$, hence $D \nmid N$ and

$$\lambda = N < \Lambda_{\alpha,\beta}.$$

□

**Corollary 6.4** ($1 < \omega < 2$)**.** *If $1 < \omega < 2$ and $D \nmid N$, then*

$$\lambda \geq \Lambda_{\alpha,\beta}.$$

*Proof.* $\lfloor \omega\alpha \rfloor \geq \alpha$ and $\lfloor \omega\beta \rfloor \geq \beta$, so

$$N \geq \alpha + \beta + 1 = \Lambda_{\alpha,\beta}.$$

Since $D \nmid N$, we have $\lambda = N \geq \Lambda_{\alpha,\beta}$. □

*Remark* 6.5. The condition $D \nmid N$ in Corollary 6.4 cannot be dropped. For $\alpha = 2$, $\beta = 1$, $\omega = 3/2$, one has

$$N = \lfloor 3 \rfloor + \lfloor 3/2 \rfloor + 1 = 5 = D.$$

Since $D \mid N$, we get $\lambda = N/D = 1$, which is less than $\Lambda_{2,1} = 4$.

**Corollary 6.6** ($\omega > 2$)**.** *If $\omega > 2$ and $D \nmid N$, then*

$$\lambda > \Lambda_{\alpha,\beta}.$$

*Proof.* $\lfloor \omega\alpha \rfloor \geq 2\alpha$ and $\lfloor \omega\beta \rfloor \geq 2\beta$, so

$$N \geq 2\alpha + 2\beta + 1 > \Lambda_{\alpha,\beta}.$$

Since $D \nmid N$, $\lambda = N > \Lambda_{\alpha,\beta}$. □

*Remark* 6.7. In both corollaries above, the degenerate case $D \mid N$ can produce small periods: $\lambda = N/D$ may be much less than $\Lambda_{\alpha,\beta}$. This occurs when the window width is tuned so that the number of tracks is an exact multiple of $D = \alpha^2 + \beta^2$.

# 7 Machine-Checked Formalisation

The algebraic core of the proof has been formalised in Lean 4 [7] using the Mathlib library [5]. The formalisation comprises approximately 1,300 lines of verified code with no `sorry` axioms, and is available in the companion repository [4] under `Lean/CutAndProject/`.

## 7.1 Architecture

The proof is structured in two layers. The first layer proves Theorem 3.1 from four abstract axioms collected in a typeclass `GeometricProjection`:

1. $N > 0$ (`N_pos`)
2. The difference sequence has period $N$ (`period_N`)
3. If $D \mid N$, the minimal period is $N/D$ (`period_degenerate`)
4. Every period $L$ induces a residue-preserving translation by some $\sigma$ with $\sigma N = LD$ (`sigma_of_period`)

Given these axioms, the generic minimality argument (`generic_minimality`, corresponding to Section 4.5) and the case dispatch (`main_theorem`, corresponding to Theorem 3.1) are proved purely algebraically.

The second layer constructs a concrete instance of this typeclass by defining the sorted multiset and difference sequence explicitly. Each axiom is discharged by a separate lemma (`N_pos_concrete`, `period_N_concrete`, `period_degenerate_concrete`, `sigma_of_period_concrete`).

## 7.2 Correspondence with the paper

Table 1 gives the correspondence between the main results of this paper and the Lean 4 declarations in `Basic.lean`.

| **Paper** | **Lean 4 name** | **Line** |
|---|---|---|
| Lemma 4.1 ($\gcd(\alpha, D) = 1$) | `coprime_alpha_D` | 11 |
| Lemma 4.1 ($\gcd(\beta, D) = 1$) | `coprime_beta_D` | 20 |
| Corollary 4.2 (residue bijection) | `residue_bijection` | 43 |
| Lemma 4.3 (non-uniform distribution) | `non_uniform_residue_distribution` | 140 |
| Uniform distribution ($D \mid N$) | `uniform_residue_distribution` | 215 |
| Heavy set is cyclic interval | `heavy_set_is_cyclic_interval` | 269 |
| Trivial stabiliser | `cyclic_interval_stabilizer_trivial` | 334 |
| Section 4.5 (generic minimality) | `generic_minimality` | 388 |
| Theorem 5.1 ($D \mid N$ case) | `period_degenerate_concrete` | 1233 |
| $\sigma N = LD$ from periodicity | `sigma_of_period_concrete` | 1206 |
| Theorem 3.1 (main result) | `main_theorem` | 1283 |

Table 1: Correspondence between paper results and Lean 4 declarations.

## 7.3 Scope of the formalisation

The formalised proof covers the algebraic and combinatorial core: coprimality, residue distribution, cyclic interval structure, the trivial stabiliser lemma, the generic minimality argument, the degenerate case, and the final case dispatch.

The geometric construction, namely defining the cut-and-project set, projecting lattice points, and sorting the multiset of $\tilde{p}$-values, is modelled abstractly via the `GeometricProjection` typeclass rather than formalised from first principles. Concretely, the sorted multiset is defined combinatorially as

$$\tilde{p}_s^{(i)} := V(i \bmod N) + \lfloor i/N \rfloor \cdot D,$$

where $V$ is the quantile function of the cumulative residue distribution. The bridge lemma `count_hits_eq_sorted_count` verifies that this construction is consistent with the residue counting function.

# 8 Related Literature

## 8.1 The three-distance theorem

The three-distance theorem of Steinhaus, Sós, Surányi, and Świerczkowski states that placing $n$ points $\{k\theta\}$ on a unit circle produces at most three distinct gap lengths [11, 2]. More recent treatments include the lattice proof by Marklof and Strömbergsson [6].

This theorem is related in spirit but differs from Theorem 3.1 in two essential ways:

- It concerns rotations on a circle rather than lattice projections through a strip of finite width.
- It characterises the number of distinct gap lengths, not the minimal period length of a gap sequence.

## 8.2 Christoffel words and Sturmian sequences

The lower Christoffel word of slope $p/q$ with $\gcd(p, q) = 1$ is a binary word of length $p + q$, and the mechanical Sturmian sequence of rational slope $p/q$ has period $p + q$ [1].

The difference from Theorem 3.1 is significant:

1. Christoffel words have period $p + q$, whereas at $\omega = 1$ our multiset construction gives $\alpha + \beta + 1$.
2. The Sturmian framework describes symbolic codings, not the projected multiset gap sequence considered here.
3. The present result concerns distances between consecutive projected multiset entries, including zero gaps when projected values coincide.

## 8.3 Cut-and-project sets and model sets

The mainstream literature on model sets [9, 8, 3] focuses largely on irrational slopes and aperiodic order. Rational slopes yield periodic structures and are therefore often less central from that perspective.

The author is not aware of a reference that states the above formula for the minimal period of the multiset gap sequence associated with a rational one-dimensional cut-and-project strip. The result is elementary once the appropriate residue-class description is introduced, but it appears useful to record explicitly.

# 9 Discussion

## 9.1 Why the formula splits into two cases

The key structural distinction is whether the $N$ tracks, viewed as arithmetic progressions modulo $D$, distribute uniformly or not:

- When $D \nmid N$, the tracks distribute non-uniformly among the $D$ residue classes. The heavy residue classes form a proper cyclic interval in the natural residue parametrisation, whose only translational symmetry is the identity. This gives $\lambda = N$.

- When $D \mid N$, each residue class is hit exactly $N/D$ times. Every integer is a projected value with multiplicity $N/D$, collapsing the gap structure to a simple pattern of period $N/D$.

### 9.2 Discarding multiplicities

The main theorem is stated for the projected multiset. If projected points are instead identified, the period may decrease. The clearest example is the degenerate case $D \mid N$. There every integer $\tilde{p}$ occurs with multiplicity $N/D$. After discarding multiplicities, the projected set is simply $\mathbb{Z}$ in the $\tilde{p}$-coordinate, so the ordinary set-valued gap sequence has constant gap and period 1.

Thus the branch $\lambda = N/D$ is a genuine multiset phenomenon. A complete formula for the set-valued gap sequence, with multiplicities discarded in all cases, is a natural companion question.

### 9.3 Open problems

1. **Dependence on the window position.** The present paper considers a strip in the specific position used in (1). If one translates the window transversally, the set of admissible invariant values changes by a shifted interval, but boundary effects may alter the interleaving pattern. It would be interesting to determine whether the same period formula persists for arbitrary window offsets.

2. **Set-valued gap sequence.** Determine the exact minimal period when projected points are counted without multiplicity. The degenerate case $D \mid N$ collapses to period 1, but the behaviour in the non-uniform case should be described explicitly.

3. **Higher dimensions.** The cut-and-project construction generalises to higher-dimensional lattices. Whether analogous period formulas hold in dimension $d \geq 2$ is unexplored.

4. **Distribution of gap lengths.** For fixed $\alpha, \beta, \omega$, one may ask for a precise characterisation of the distinct gap values and their multiplicities within one minimal period.


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## References


[1] Jean Berstel, Aaron Lauve, Christophe Reutenauer, and Franco V. Saliola, *Combinatorics on Words: Christoffel Words and Repetitions in Words*, CRM Monograph Series, Vol. 27, American Mathematical Society, 2008.

[2] Valérie Berthé and Christophe Reutenauer, "On the three distance theorem," *The Mathematical Intelligencer*, 2024. DOI: 10.1007/s00283-023-10316-z.

[3] Alan Haynes, Henna Koivusalo, James Walton, and Lorenzo Sadun, "Gaps problems and frequencies of patches in cut and project sets," *Mathematical Proceedings of the Cambridge Philosophical Society*, vol. 161, no. 1, pp. 65–85, 2016.

[4] Dirk Kunert, *Repository cut-and-project*, 2025. Available at: `https://github.com/dkunert/cut-and-project`.

[5] The mathlib Community, "The Lean mathematical library," *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pp. 367–381, 2020. DOI: 10.1145/3372885.3373824.

[6] Jens Marklof and Andreas Strömbergsson, "The three gap theorem and the space of lattices," *The American Mathematical Monthly*, vol. 124, no. 8, pp. 741–745, 2017.

[7] Leonardo de Moura, Sebastian Ullrich, "The Lean 4 theorem prover and programming language," *Proceedings of the 28th International Conference on Automated Deduction (CADE-28)*, LNAI, vol. 12699, pp. 625–635, Springer, 2021. DOI: 10.1007/978-3-030-79876-5_37.

[8] Yves Meyer, *Algebraic Numbers and Harmonic Analysis*, North-Holland, Amsterdam, 1972.

[9] Marjorie Senechal, *Quasicrystals and Geometry*, Cambridge University Press, 2009. ISBN: 978-0-521-57541-6.

[10] Terence Tao, "Terence Tao on Yves Meyer's work on wavelets," video available at: `https://youtu.be/AnkinNVPjyw`, accessed December 15, 2024.

[11] Vera T. Sós, "On the distribution mod 1 of the sequence $n\alpha$," *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica*, vol. 1, pp. 127–134, 1958.